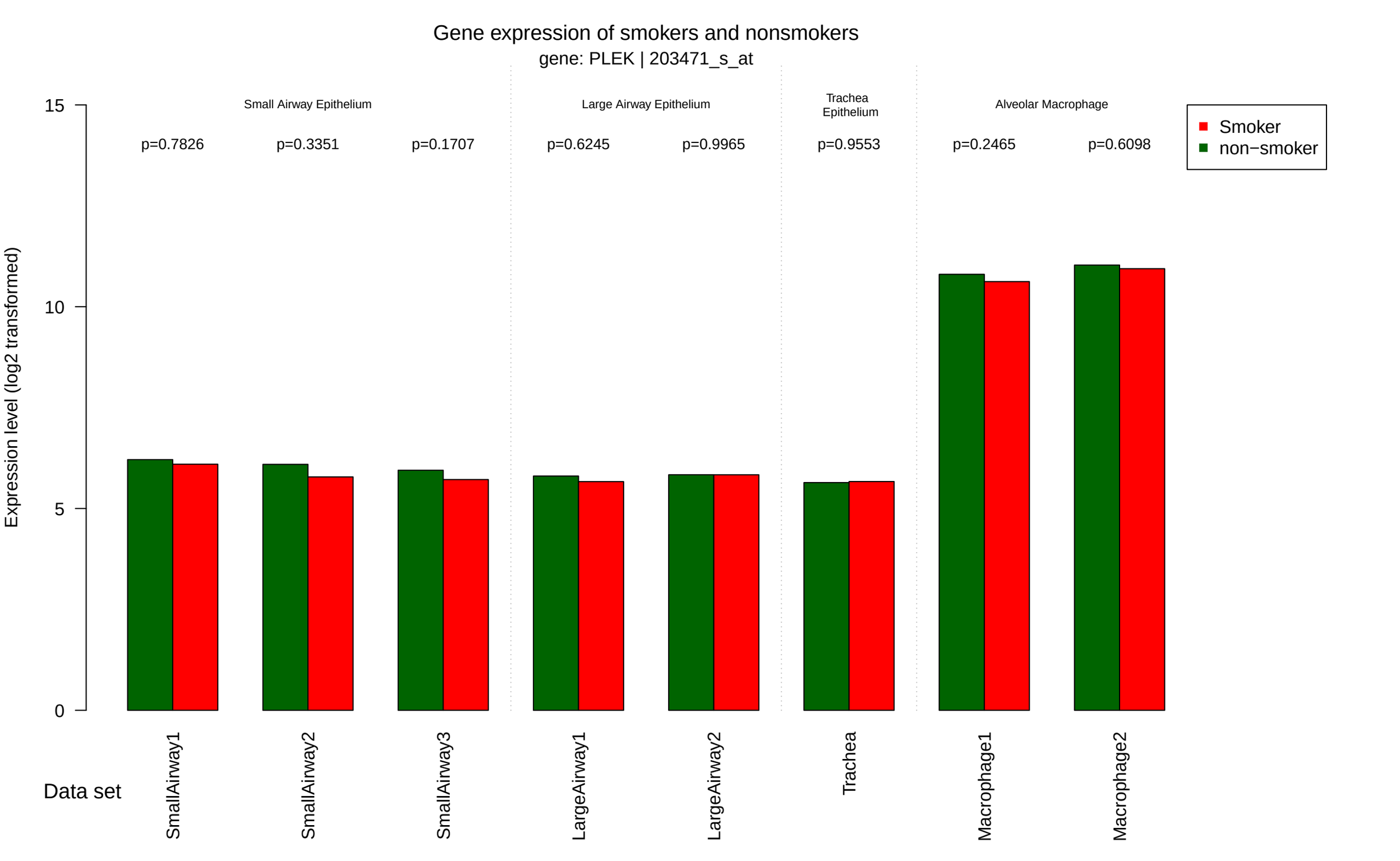
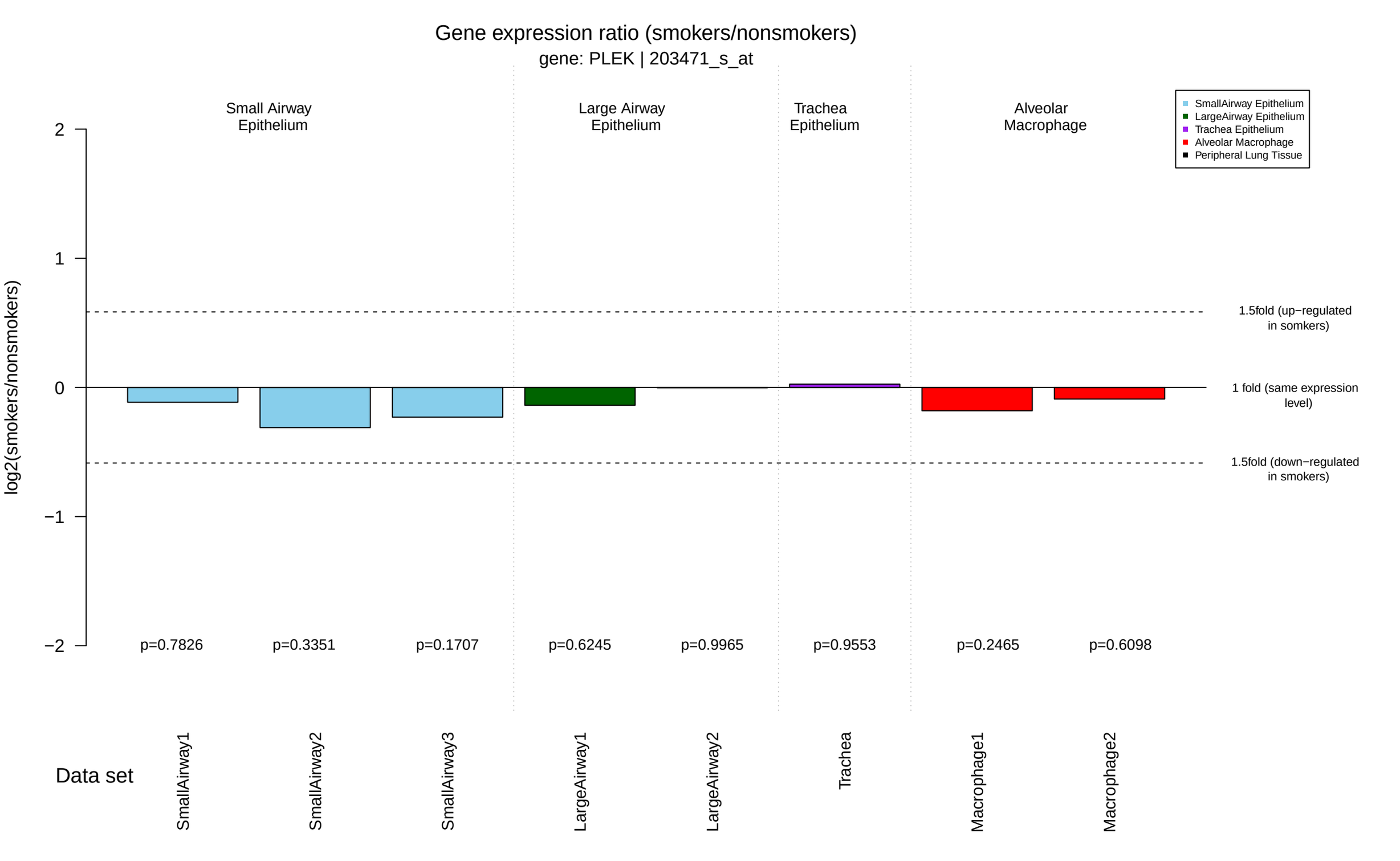
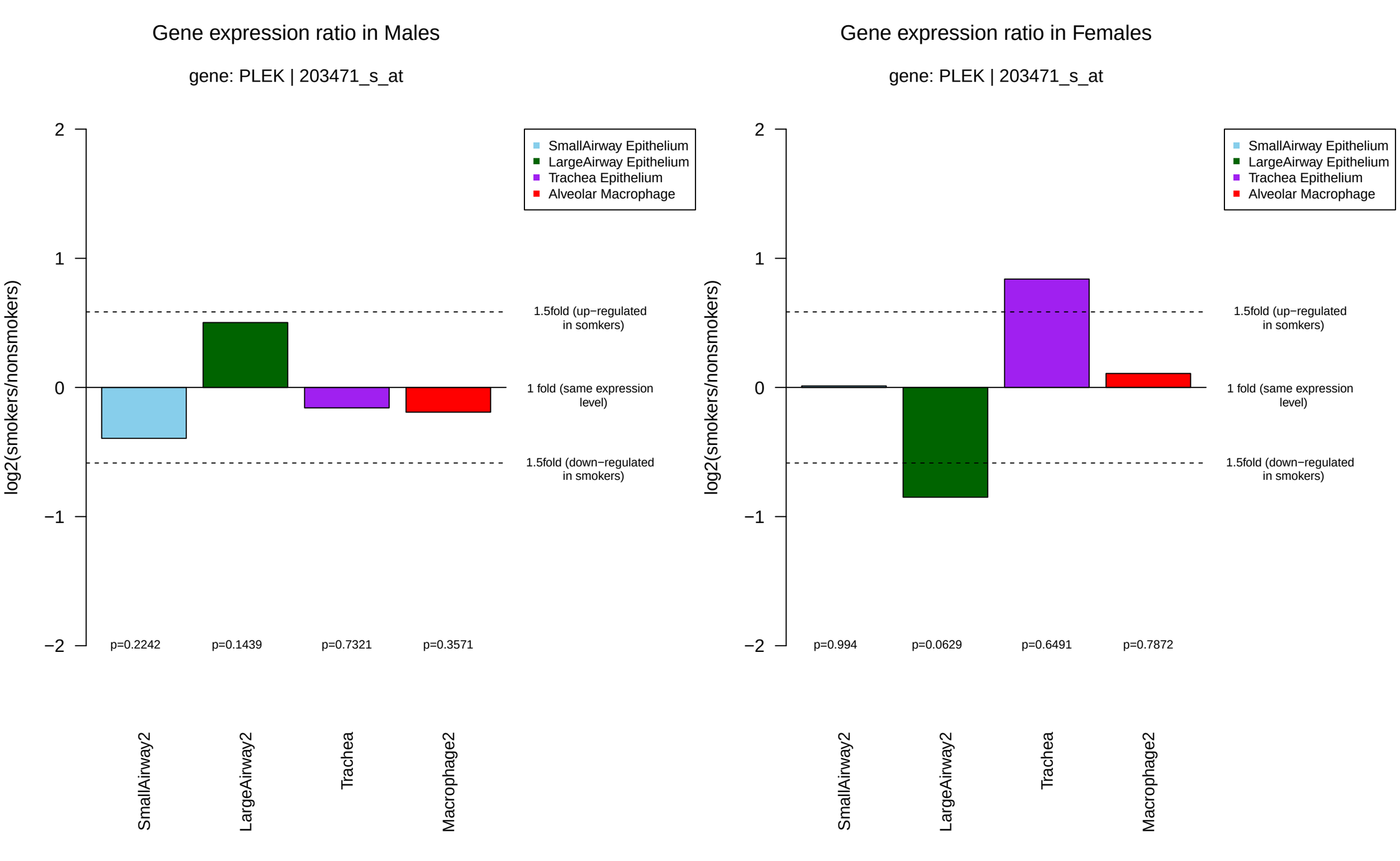
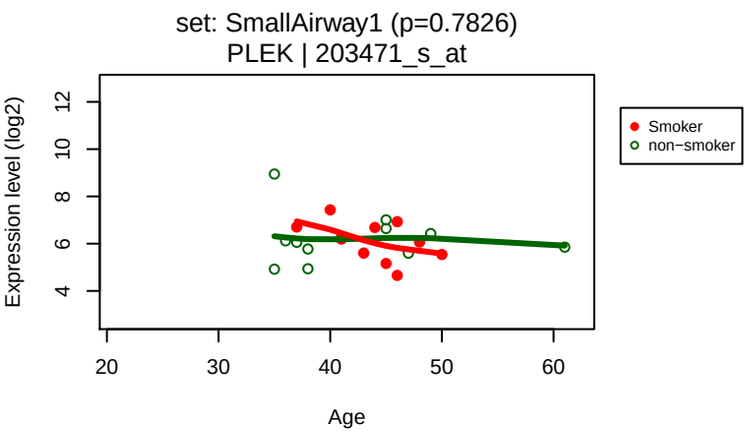
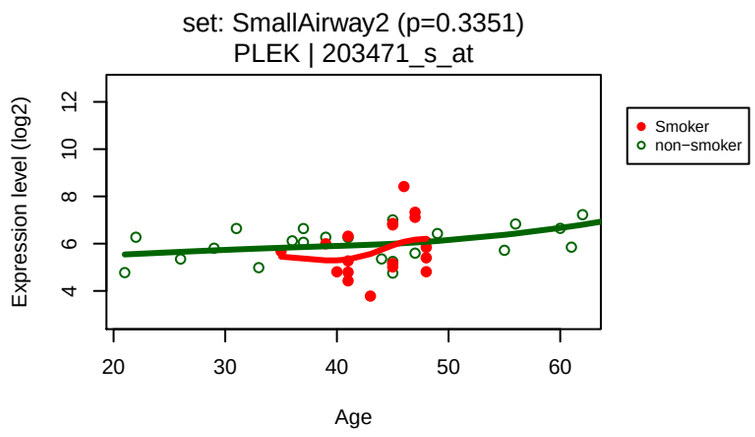
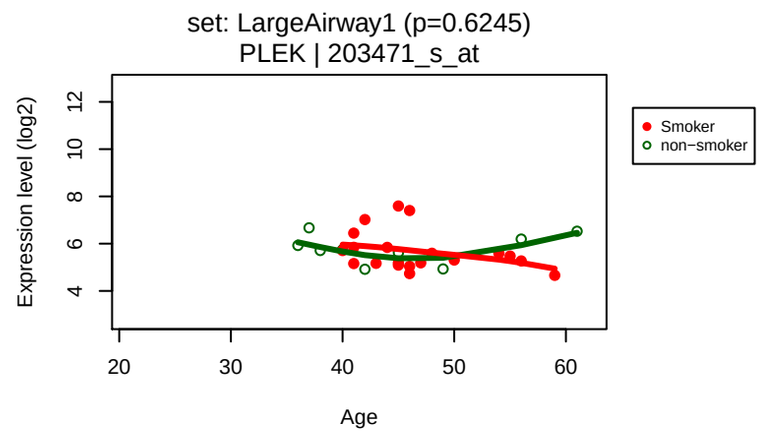

Gene expression of smokers and nonsmokers
gene: PLEK | 203471_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7826
p=0.3351
p=0.1707
p=0.6245
p=0.9965
p=0.9553
p=0.2465
p=0.6098
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

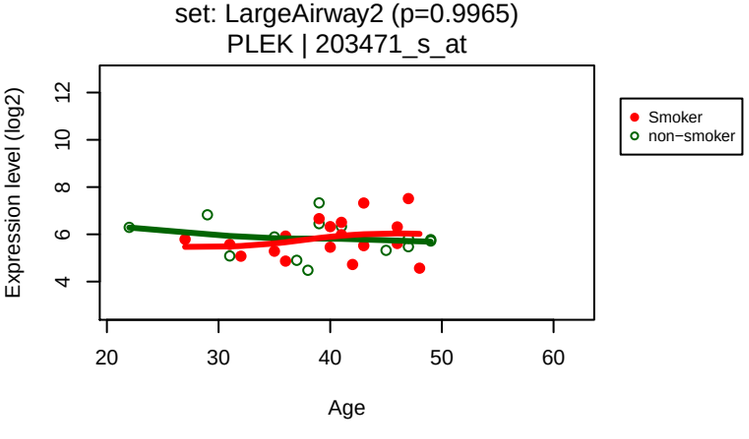

Gene expression ratio (smokers/nonsmokers)
gene: PLEK | 203471_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7826
p=0.3351
p=0.1707
p=0.6245
p=0.9965
p=0.9553
p=0.2465
p=0.6098
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

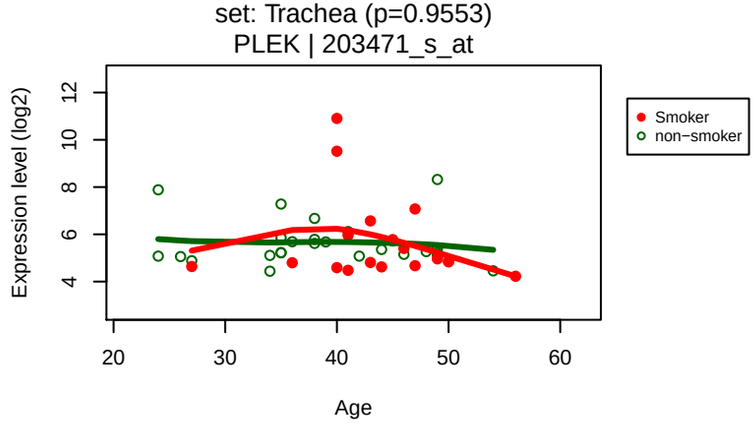

Gene expression ratio in Males
gene: PLEK | 203471_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2242
p=0.1439
p=0.7321
p=0.3571
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PLEK | 203471_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.994
p=0.0629
p=0.6491
p=0.7872
SmallAirway2
LargeAirway2
Trachea
Macrophage2

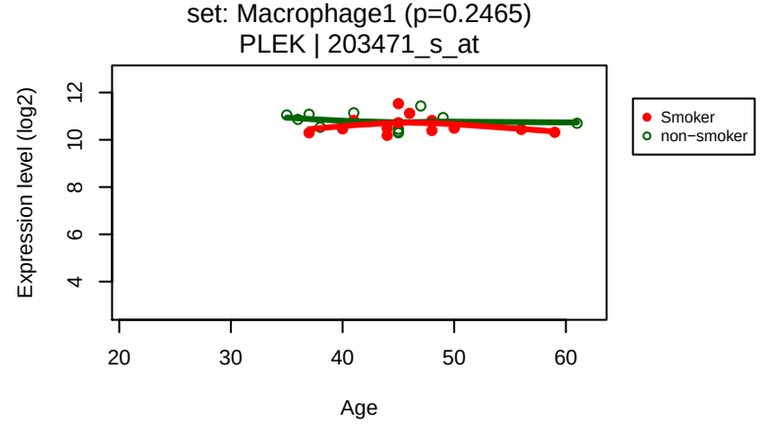

set: SmallAirway1 (p=0.7826)
PLEK | 203471_s_at
Expression level (log2)
Age
Smoker
non−smoker

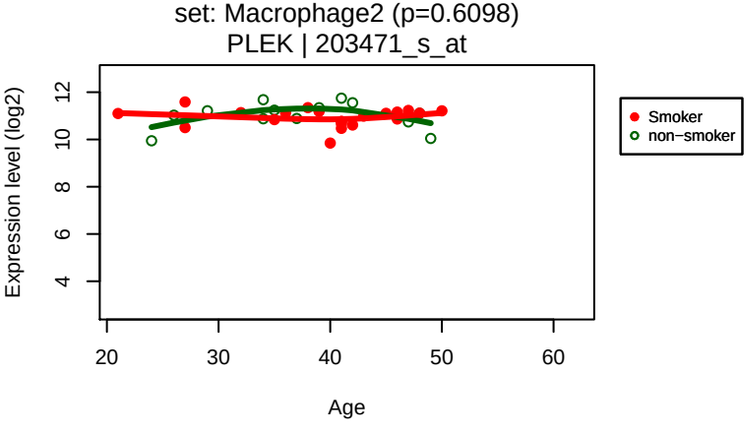

set: SmallAirway2 (p=0.3351)
PLEK | 203471_s_at
Expression level (log2)
Age
Smoker
non−smoker

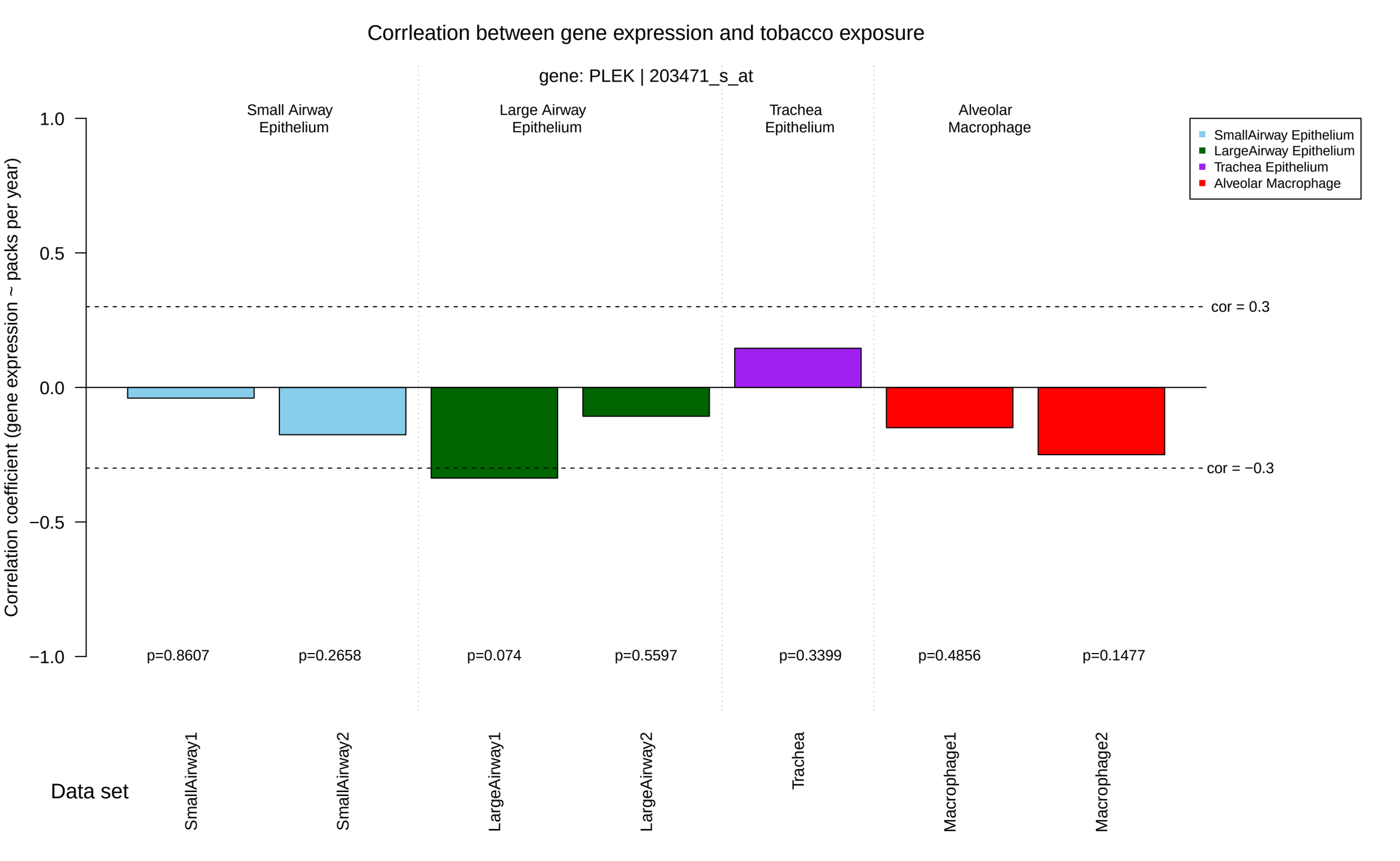

set: LargeAirway1 (p=0.6245)
PLEK | 203471_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9965)
PLEK | 203471_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9553)
PLEK | 203471_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2465)
PLEK | 203471_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6098)
PLEK | 203471_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PLEK | 203471_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8607
p=0.2658
p=0.074
p=0.5597
p=0.3399
p=0.4856
p=0.1477
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2